

(c)

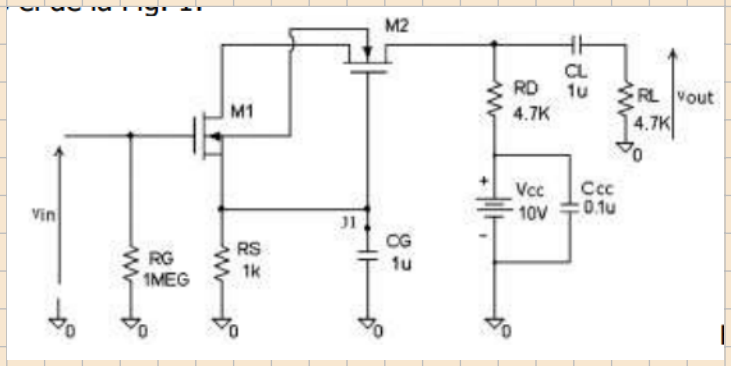
T_1: kP = 15mA/V^2 ; VTO =-1V ; W/L=1

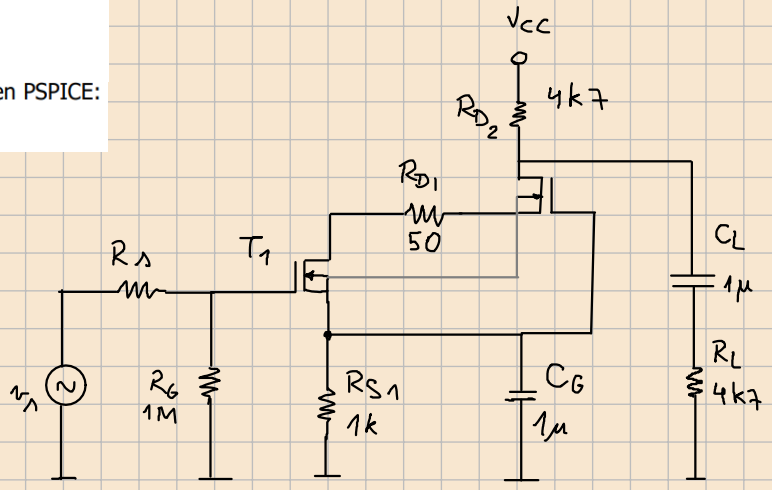
T_2: kP =200mA/V^2 ; VTO =-1V ; W/L=1

Modelo simplificado a utilizar

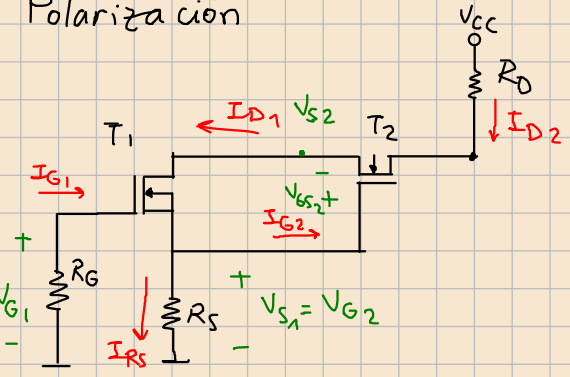
La nomenclatura de la figura 3c corresponde a la ecuación del MOSFET utilizada en PSPICE:

$I_D = (kp/2).(W/L).(V_{GS}-V_T)^2$; donde $kp=(\mu C_{ox}')=2.k'$





Polarización



- A suma T_1 y T_2 en SAT
- $I_{D_1} = I_{D2}$ e $I_{G_1} = I_{G2} = 0$
- $I_{RS} = \frac{V_{S1}}{R_S} = \frac{-V_{GS1}}{R_S}$

$V_{GS_1} = -V_{S_1}$

$V_{GS_2} = V_{S1} - V_{S2} = -V_{GS1} - V_{S2}$

$I_{D_1} = k_1 (V_{GS_1} - V_T)^2 = \frac{-V_{GS1}}{R_S} = I_{RS_2}$

★ $k_i = \frac{kP_i}{2}$

$k_1 (V_{GS_1}^2 - 2 V_{GS_1} V_T + V_T^2) + \frac{V_{GS1}}{R_S} = 0$

$k_1 V_{GS_1}^2 - V_{GS} \left(2kV_T - \frac{1}{R_S}\right) + k V_T^2 = 0$

$k_1 = 7{,}5 \frac{mA}{V^2}$ $\quad V_T = -1V$

$R_S = 1k\Omega$

$V_{GS1} = \begin{cases} \frac{-16+\sqrt{31}}{15} \approx -695{,}5\ mV & \checkmark \\ \frac{-16-\sqrt{31}}{15} \approx -1{,}438\ mV & \times \end{cases}$

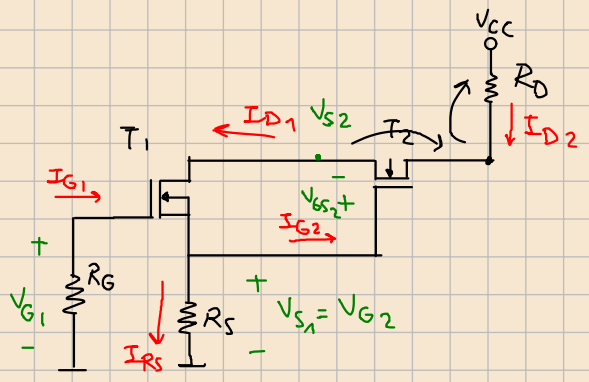
$V_{GS} > V_T$ por ser NMOS!

$I_{D_1} = k_1 (V_{GS_2} - V_T)^2 = 695{,}5\ \mu A = I_{D2}$

$I_{D2} = k_2 (V_{GS_2} - V_T)^2 \Rightarrow \pm\sqrt{\frac{I_{D2}}{K_2}} = (V_{GS_2} - V_T) \Rightarrow V_{GS_2} = V_T \pm \sqrt{\frac{I_{D2}}{k_2}}$

$V_{GS_2} = \begin{cases} -0{,}9166\ V & \checkmark \\ -1{,}0833\ V & \times \end{cases} \rightarrow V_{GS} > V_T$

$V_{GS_2} = -V_{DS_1}$



$V_{S_1} + V_{DS_1} + V_{DS_2} + I_{D_2} R_D - V_{CC} = 0$

$V_{DS_2} = V_{CC} - V_{S_1} - V_{DS_1} - I_{D_2} R_D$

$V_{DS_2} = 10\,V - 69\,mV - 917\,mV - 3,269\,V$

$V_{DS_2} = 5,12\,V$

$V_{D_2} = V_{CC} - I_{D_2} R_D$

$V_{D_2} = 6,731\,V$